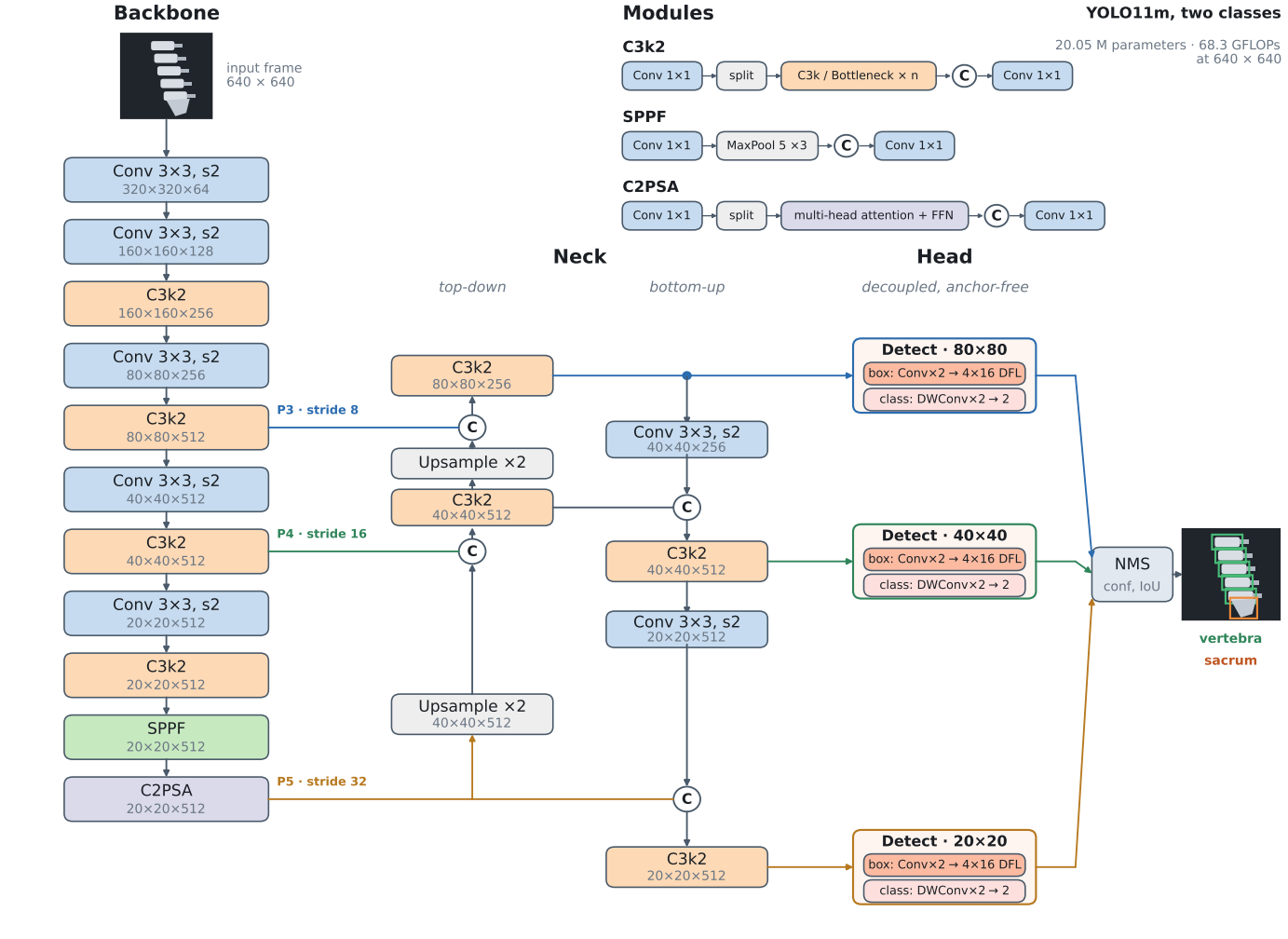
Backbone
input frame
640 × 640
Conv 3×3, s2
320×320×64
Conv 3×3, s2
160×160×128
C3k2
160×160×256
Conv 3×3, s2
80×80×256
C3k2
80×80×512
P3 · stride 8
Conv 3×3, s2
40×40×512
C3k2
40×40×512
P4 · stride 16
Conv 3×3, s2
20×20×512
C3k2
20×20×512
SPPF
20×20×512
C2PSA
20×20×512
P5 · stride 32
Modules
YOLO11m, two classes
20.05 M parameters · 68.3 GFLOPs
at 640 × 640
C3k2
Conv 1×1
split
C3k / Bottleneck × n
C
Conv 1×1
SPPF
Conv 1×1
MaxPool 5 ×3
C
Conv 1×1
C2PSA
Conv 1×1
split
multi-head attention + FFN
C
Conv 1×1
Neck
top-down
bottom-up
Head
decoupled, anchor-free
C3k2
80×80×256
C
Upsample ×2
C3k2
40×40×512
C
Upsample ×2
40×40×512
Conv 3×3, s2
40×40×256
C
C3k2
40×40×512
Conv 3×3, s2
20×20×512
C
C3k2
20×20×512
Detect · 80×80
box: Conv×2 → 4×16 DFL
class: DWConv×2 → 2
Detect · 40×40
box: Conv×2 → 4×16 DFL
class: DWConv×2 → 2
Detect · 20×20
box: Conv×2 → 4×16 DFL
class: DWConv×2 → 2
NMS
conf, IoU
vertebra
sacrum